DYLAN BUSTOS
Cedar Park, TX - 512-925-1536 - dylanantoniobustos@gmail.com - Portfolio - GitHub

SUMMARY
Software engineer focused on data-driven product development, with experience building Python and SQL applications, machine learning models, and analytics pipelines end to end—from data extraction and cleaning to model deployment–ready code and visualization. I have delivered projects that predict hotel booking cancellations, forecast visa approval outcomes, and segment stock market behavior, consistently translating model outputs into clear business recommendations. I am seeking entry-level software engineering or data science roles where I can ship production-quality code, collaborate with cross-functional teams, and use ML and analytics to drive measurable impact on revenue, retention, and customer experience.

TECHNICAL SKILLS
Languages (strong): Python, SQL
Languages (working knowledge): Java, C++, JavaScript
Data & ML: Pandas, NumPy, scikit-learn, statistical modeling, supervised learning (classification), unsupervised learning (clustering), feature engineering, model evaluation
Visualization & Analytics: Matplotlib, Seaborn, Tableau, Excel
Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, MySQL
Software Engineering: Data structures and algorithms, object-oriented programming, database systems, debugging and iterative model refinement

EXPERIENCE

EasyVisa – ML Approval Predictor | Python, scikit-learn, Pandas, NumPy
Remote | 2025

- Built an end-to-end classification pipeline to predict visa approvals using ensemble methods (bagging and boosting), including data preprocessing, feature engineering, and hyperparameter tuning.
- Achieved 75–78% accuracy on held-out data and identified the most important drivers of approval to support policy and staffing decisions.
- Communicated model results and trade-offs to non-technical stakeholders through clear visualizations and narrative insights, enabling data-informed decision-making.

INN Hotels – Booking Cancellation Risk Model | Python, scikit-learn, SQL
Remote | 2025

- Developed logistic regression and decision tree models to predict hotel booking cancellations using transactional and customer-level features.
- Identified top drivers of cancellation risk and quantified their impact, informing refund policies and revenue management strategies for high-risk bookings.
- Wrote reusable SQL queries and Python scripts to automate model-ready data extraction and ongoing performance monitoring.

Trade & Ahead – Market Behavior Clustering | Python, scikit-learn, NumPy, Matplotlib
Remote | 2025

- Implemented K-Means and hierarchical clustering to segment stock market behavior based on price and volume features, including scaling and cluster selection.
- Evaluated clustering quality with cophenetic correlation, achieving a score of 0.94 using Euclidean distance and average linkage, and used profiles to describe distinct trading regimes.
- Produced visual reports summarizing cluster characteristics to support data-driven strategy exploration.

New Wheels – Customer Retention Analytics | SQL, Tableau
Remote | 2025

- Designed complex SQL queries to analyze 994 yearly customer reviews and retention metrics, transforming raw data into structured reporting tables.
- Built an interactive Tableau dashboard to monitor churn risk, satisfaction drivers, and segment-level performance for quarterly business reviews.
- Automated quarterly reporting workflows, reducing manual analysis time and making retention insights accessible to non-technical stakeholders.

FoodHub – Customer Order Behavior Analysis | Python, Pandas, NumPy, Matplotlib, Seaborn
Remote | 2025

- Conducted exploratory data analysis on food delivery orders to uncover customer ordering patterns, order frequency, and basket composition.
- Identified behavioral segments and promotional opportunities to guide targeted marketing campaigns.
- Created clear visualizations and summary metrics to support experimentation and promotion design.

EDUCATION

The University of Texas at Austin – Post Graduate Program, Data Science and Business Analytics (GPA: 4.0)
Austin, TX | Aug 2025
Relevant Coursework: Machine Learning, Probability & Statistics, Data Visualization, Business Analytics

The University of Texas at Dallas – Bachelor of Science, Computer Science
Richardson, TX | May 2024
Relevant Coursework: Data Structures and Algorithms, Database Systems, Machine Learning, Software Engineering, Operating Systems

LEADERSHIP & INVOLVEMENT

Honor Board Member, Delta Tau Delta | 2020–2023

- Investigated and resolved chapter bylaw violations, applying structured decision-making and documentation to strengthen accountability and compliance.
- Partnered with chapter leadership to communicate outcomes and maintain chapter standards.

Director of Risk Management, Delta Tau Delta | 2022–2023

- Led risk management initiatives and compliance reviews across chapter operations, assessing and mitigating organizational risk.
- Coordinated a team of Honor Board members and advised executive leadership on policy updates to improve safety and operational oversight.